

The $m = 2$ decoder moment in terms of Q and $Q^\dagger$

Companion to `progress_2026-09-29.pdf`, Fig. 1 and §3.2

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1 Ingredients

Supercharge. One-flavor $\mathcal{N}=2$ SYK on N' complex fermions, with p odd ($p = 3$):

$$Q = \sum_{|I|=p} C_I \psi_I, \quad Q^\dagger = \sum_{|I|=p} \bar{C}_I \psi_I^\dagger, \quad \psi_I = \psi_{i_1} \cdots \psi_{i_p} \ (i_1 < \cdots < i_p), \quad (1)$$

with complex Gaussian couplings. Their only nonzero second moment is

$$\mathbb{E}[C_I \bar{C}_J] = \delta_{IJ}, \quad \mathbb{E}[C_I C_J] = \mathbb{E}[\bar{C}_I \bar{C}_J] = 0. \quad (2)$$

(The physical variance only rescales β below, so we set it to 1.) A **chord** is one Wick pair $C_I \bar{C}_I$. It always joins a Q to a $Q^\dagger$. Q - Q and $Q^\dagger$ - $Q^\dagger$ pairs vanish by (2).

The BPS projector contains both Q and $Q^\dagger$. With $H = \{Q, Q^\dagger\} = QQ^\dagger + Q^\dagger Q \geq 0$,

$$P = \lim_{\beta \rightarrow \infty} e^{-\beta H} \Big|_{\Lambda^F} = \lim_{\beta \rightarrow \infty} \sum_{a \geq 0} \frac{(-\beta)^a}{a!} H^a. \quad (3)$$

Since $Q^2 = 0$ (odd p), $H^a = (QQ^\dagger)^a + (Q^\dagger Q)^a$ for $a \geq 1$. *Every word in P alternates and contains equally many Q 's and $Q^\dagger$'s.* This is why chords can live inside P .

Parity probe. $U = (-1)^{n_i}$ gives $U\psi_i U = -\psi_i$ and $U\psi_j U = \psi_j$ for $j \neq i$, so

$$U\psi_I U = s_I \psi_I, \quad s_I \equiv (-1)^{[i \in I]}, \quad UQU = \sum_I s_I C_I \psi_I, \quad UQ^\dagger U = \sum_I s_I \bar{C}_I \psi_I^\dagger. \quad (4)$$

Hence $P' \equiv UPU$ is the same function of the *sign-flipped* couplings $C_I \rightarrow s_I C_I$.

2 Where the U 's go

Three P 's become two arcs. $(PUP)^2 = PUPUP$ has three P 's, but under the trace cyclicity and $P^2 = P$ give

$$\text{Tr}[(PUP)^2] = \text{Tr}[PUPUP] = \text{Tr}[PPUPU] = \text{Tr}[PUPU]. \quad (5)$$

On the chord circle, the first and last P are adjacent segments of a single arc.

Moving U does not make it disappear for free. $UP = (UPU)U = P'U$, so

$$\text{Tr}[PUPU] = \text{Tr}[P(UPU)] = \text{Tr}[PP']. \quad (6)$$

The two U 's annihilate, *but the P they passed is now P'* . By (4), every supercharge in that arc carries a sign s_I .

3 Explicit expansion

Expand both projectors:

$$\text{Tr}[e^{-\beta_1 H} U e^{-\beta_2 H} U] = \sum_{a,b \geq 0} \frac{(-\beta_1)^a (-\beta_2)^b}{a! b!} \text{Tr}[H^a H'^b], \quad H' = UHU. \quad (7)$$

Lowest nontrivial order: $a = b = 1$

Write

$$H = \sum_{I,J} C_I \bar{C}_J A_{IJ}, \quad A_{IJ} \equiv \psi_I \psi_J^\dagger + \psi_J^\dagger \psi_I, \quad H' = UHU = \sum_{K,L} s_K s_L C_K \bar{C}_L A_{KL}. \quad (8)$$

Then

$$\text{Tr}[H U H U] = \text{Tr}[H H'] = \sum_{I,J,K,L} s_K s_L C_I \bar{C}_J C_K \bar{C}_L \text{Tr}[A_{IJ} A_{KL}]. \quad (9)$$

By (2) the Gaussian average has exactly two Wick pairings:

$$\mathbb{E}[C_I \bar{C}_J C_K \bar{C}_L] = \underbrace{\delta_{IJ} \delta_{KL}}_{(i)} + \underbrace{\delta_{IL} \delta_{JK}}_{(ii)}. \quad (10)$$

(i) No joining chord. Q pairs with $Q^\dagger$ inside H , and Q' with $Q'^\dagger$ inside H' :

$$(i) = \sum_{I,K} \underbrace{s_K^2}_{=1} \text{Tr}[A_{II} A_{KK}] = \sum_{I,K} \text{Tr}[A_{II} A_{KK}]. \quad (11)$$

The chord inside the P' -arc carries s_K at *both* ends, so the signs cancel. Weight 1.

(ii) Two joining chords. Chord I joins the Q in H to the $Q'^\dagger$ in H' . Chord K joins the $Q^\dagger$ in H to the Q' in H' :

$$(ii) = \sum_{I,K} s_I s_K \text{Tr}[A_{IK} A_{KI}]. \quad (12)$$

Each joining chord carries *one* sign, from its end in the P' -arc.

No single joining chord. H holds one Q and one $Q^\dagger$. If its Q pairs across, its $Q^\dagger$ has no partner left except across.

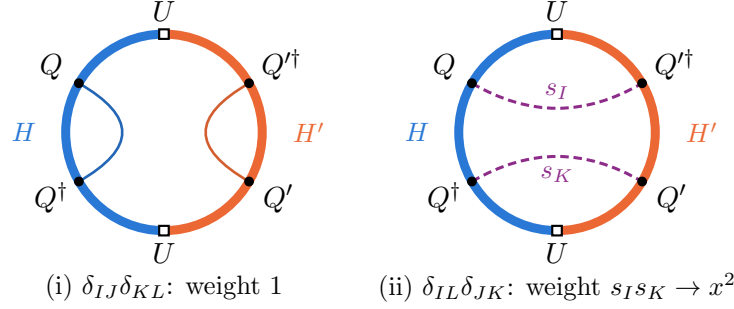


Figure 1: The two Wick pairings of the term $\text{Tr}[Q Q^\dagger U Q' Q'^\dagger U]$ in $\text{Tr}[HUHU]$ (read counterclockwise from the top). Blue: P -arc. Orange: P' -arc. Squares: U . A chord whose endpoints lie on opposite sides of a U carries one sign s_I . The pairing $Q-Q'$, $Q^\dagger-Q'^\dagger$ would give a single crossing diagram, but it vanishes by (2).

4 All orders

In $\text{Tr}[H^a H'^b]$, the summand for fixed indices is the same fermion trace as in $\text{Tr}[H^{a+b}]$. The only difference is a factor s_I on each primed supercharge. After Wick contraction:

- a chord inside the P -arc carries no sign;
- a chord inside the P' -arc carries $s_I^2 = 1$;
- a chord joining the arcs carries exactly one s_I .

Hence

$$\text{Tr}[H^a H'^b] = \sum_{\text{pairings } \pi} \sum_{\text{indices}} \left(\prod_{c \in \text{joining}(\pi)} s_{I_c} \right) \times [\text{summand of } \text{Tr}(H^{a+b}) \text{ for } \pi]. \quad (13)$$

All crossing weights (q -factors) sit in the fermion trace and are unchanged by U . The probe contributes only the signs.

Joining chords come in pairs. The P -arc word has a Q 's and a $Q^\dagger$'s. Each internal chord uses one of each, since $Q-Q$ pairs vanish. The ends left over to join across are therefore balanced: n Q -ends and n $Q^\dagger$ -ends, i.e. $2n$ joining chords.

Averaging the signs. For a uniformly random p -subset, $\text{Pr}[i \in I] = p/N'$, so

$$\overline{s_I} = 1 - \frac{2p}{N'} \equiv x \quad (\text{exact, one chord}). \quad (14)$$

If the $2n$ signs are replaced independently by their mean, a diagram with $2n$ joining chords is weighted by x^{2n} . Taking $\beta_{1,2} \rightarrow \infty$ and resumming gives

$$T_2 = \tau(PP') = \sum_{n \geq 0} P_n x^{2n}, \quad (15)$$

with P_n the probability that the zero-temperature wormhole has n chords (BLY eqs. 4.5–4.8 with $\lambda = 2p^2/N'$, $\Delta = 1/p$; in that limit $x \simeq e^{-2p/N'} = q^{1/p}$).

Caveat (finite N). Replacing each s_{I_c} by x *independently* is the double-scaled approximation. At finite N , the fermion trace (e.g. $\text{Tr}[A_{IK}A_{KI}]$) depends on how I and K overlap, including whether i lies in both. So the weighted average of $s_I s_K$ is not exactly x^2 . The exact finite- N statement for T_2 is the size sum rule of §4 of the progress report, which needs no chord factorization.